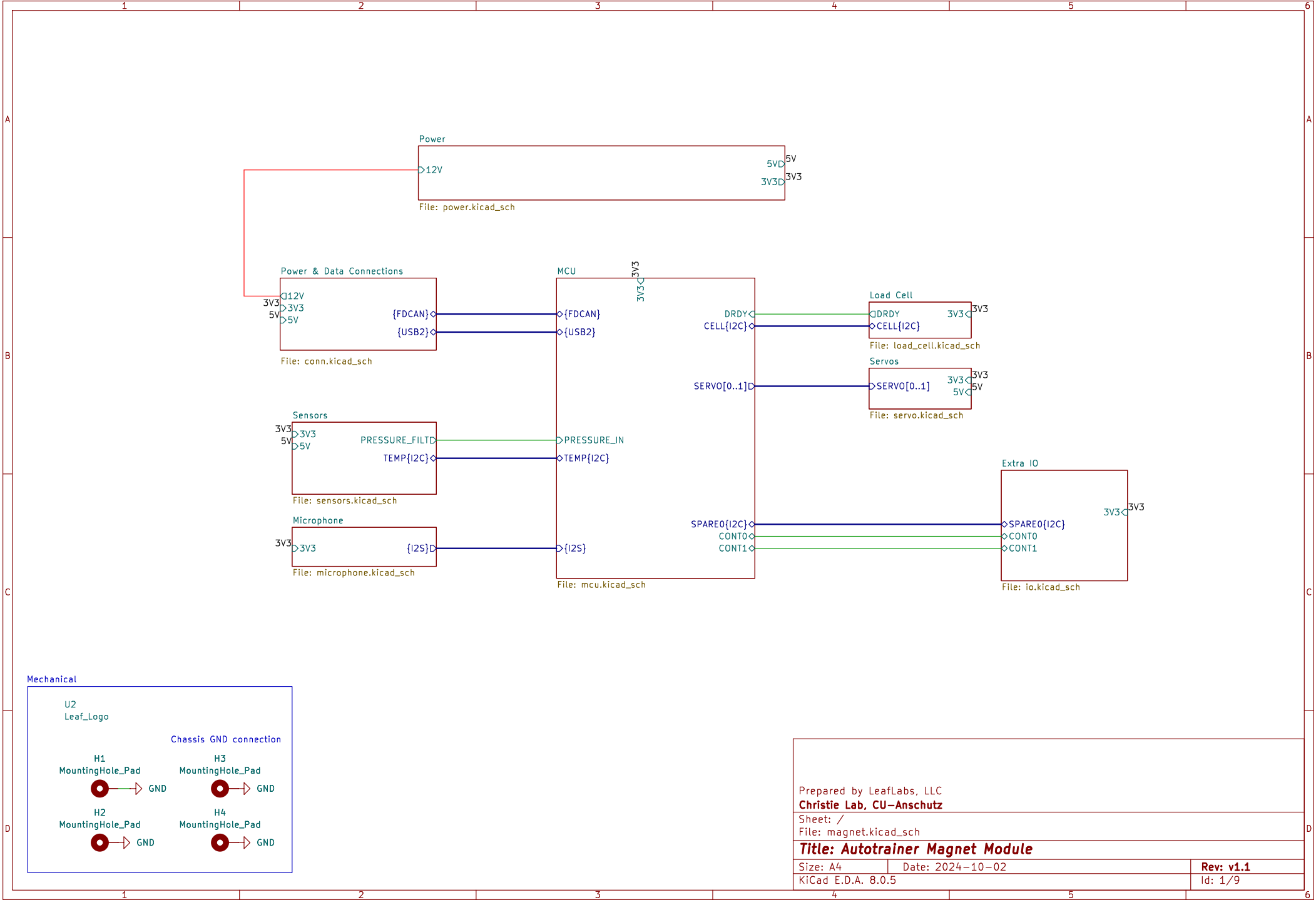
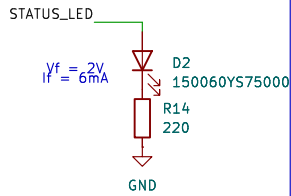
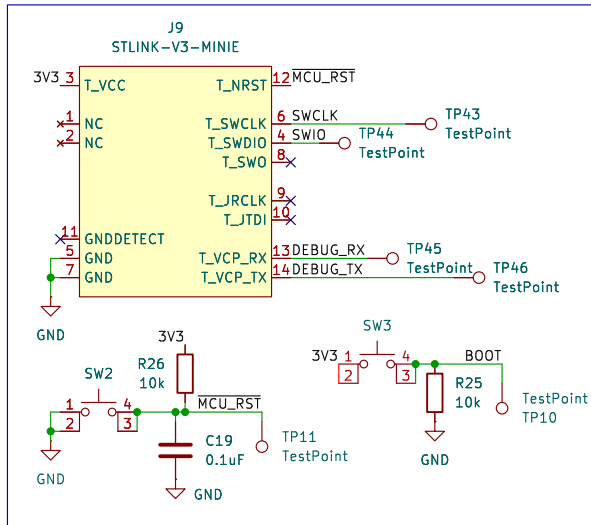




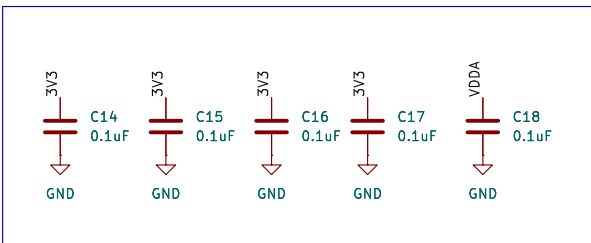
Status LED



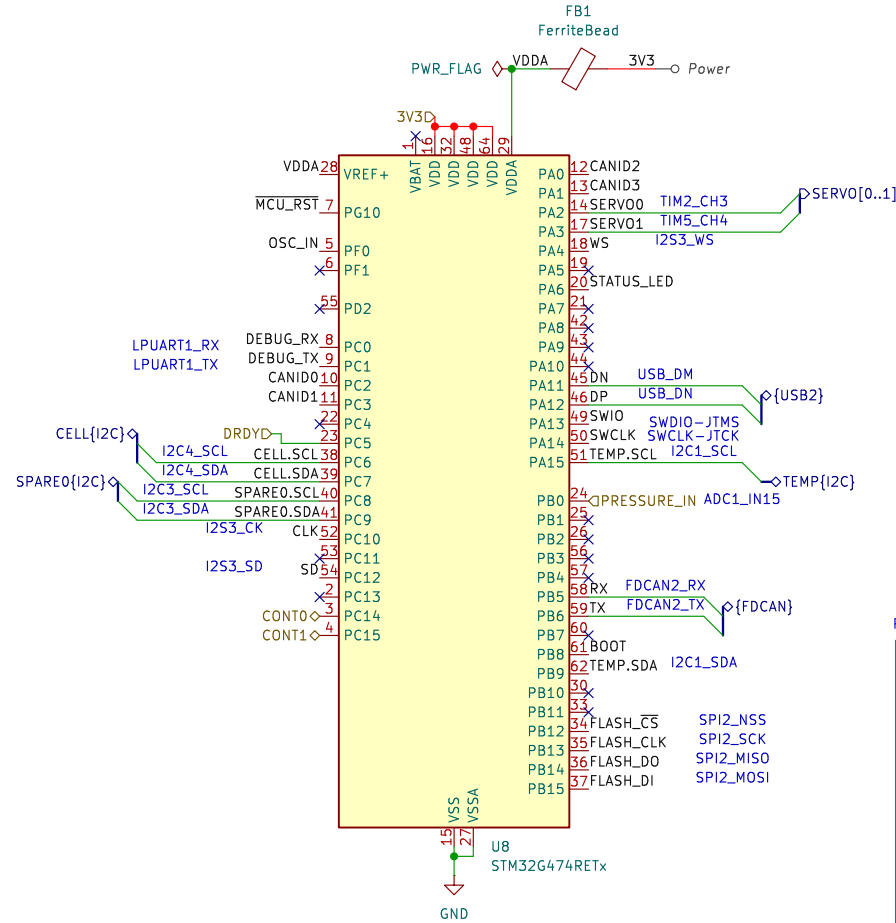
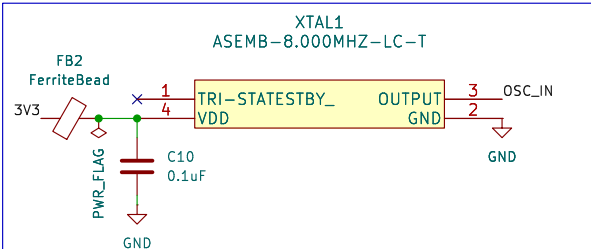
Debug



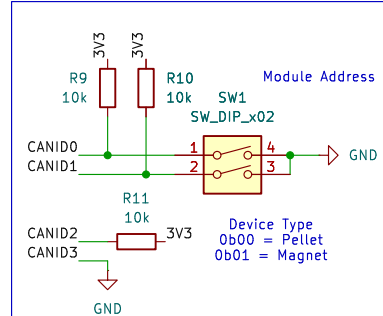
Decoupling Capacitor Central



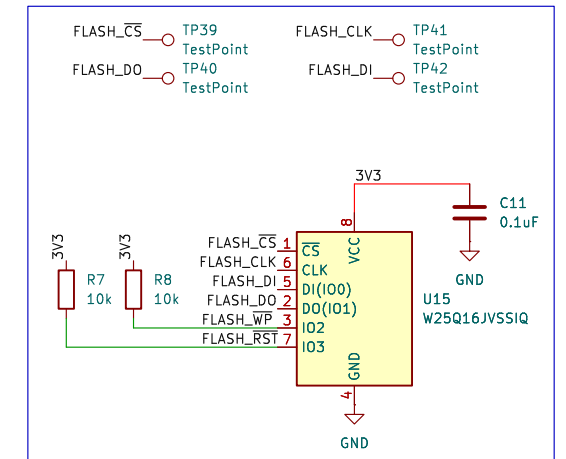
External clock



CAN Addressing



Programming Flash



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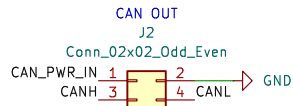
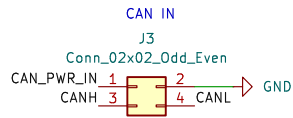
Sheet: /MCU/
File: mcu.kicad_sch

Title: Autotrainer Magnet Module – MCU

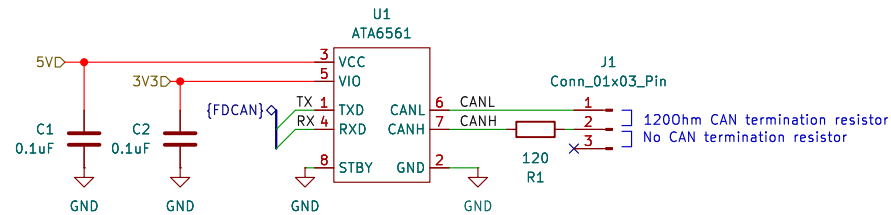
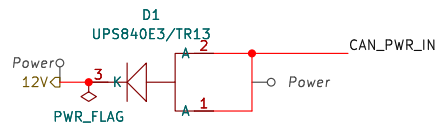
Size: A4 Date: 2024-10-02
KiCad E.D.A. 8.0.5

Rev: v1.1
Id: 2/9

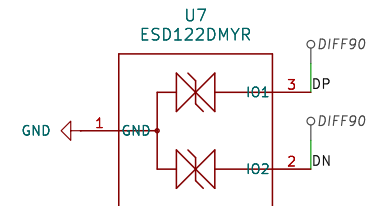
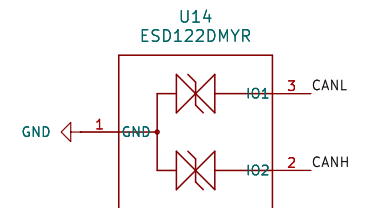
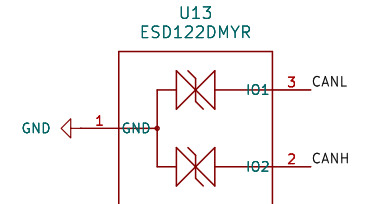
CAN Connections



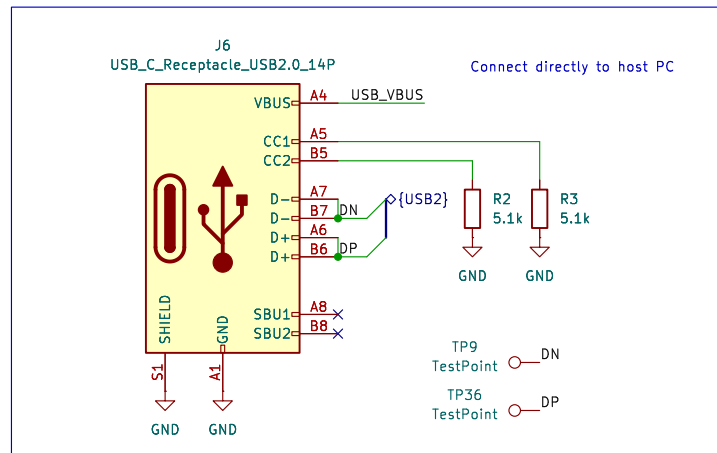
Reverse polarity & backpower protection



ESD Protection



USB Connector



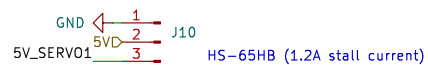
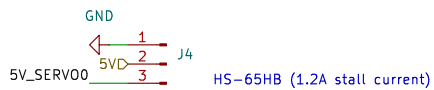
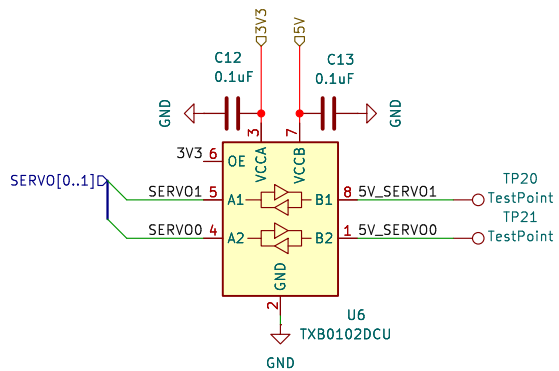
- TP32 TestPoint ○ CAN_PWR_IN
- TP33 TestPoint ○ CANH
- TP34 TestPoint ○ CANL
- TP35 TestPoint ○ GND
- TP37 TestPoint ○ TX
- TP38 TestPoint ○ RX

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Sheet: /Power & Data Connections/
File: conn.kicad_sch

Title: Autotrainer Magnet Module – Power & Data Conn

Size: A4 Date: 2024-10-02 Rev: v1.1
KiCad E.D.A. 8.0.5 Id: 3/9

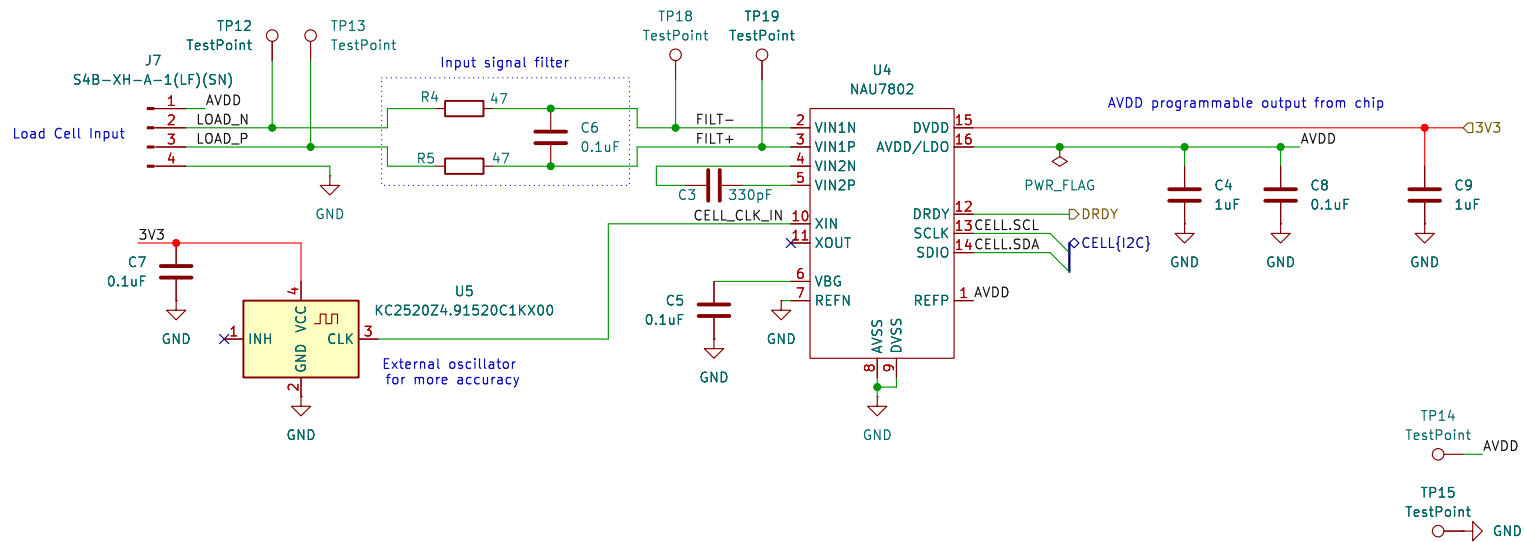


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Sheet: /Servos/
File: servo.kicad_sch

Title: Autotrainer Magnet Module – Servos

Size: A4	Date: 2024-10-02	Rev: v1.1
KiCad E.D.A. 8.0.5		Id: 4/9



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Sheet: /Load Cell/
 File: load_cell.kicad_sch

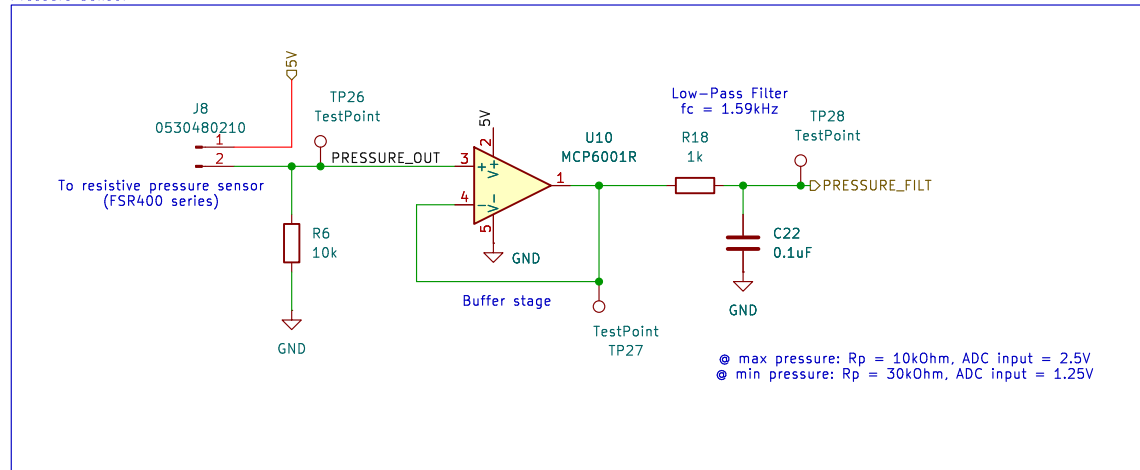
Title: Autotrainer Magnet Module – Load Cell

Size: A4 Date: 2024-10-02

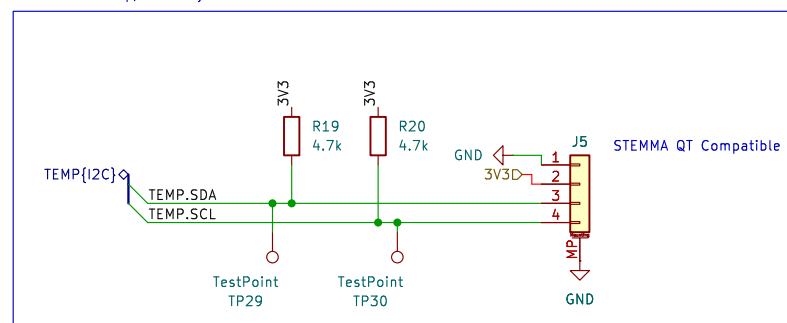
KiCad E.D.A. 8.0.5

Rev: v1.1
 Id: 6/9

Pressure Sensor



To external temp/humidity sensor



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Sheet: /Sensors/
 File: sensors.kicad_sch

Title: Autotrainer Magnet Module – Sensors

Size: A4

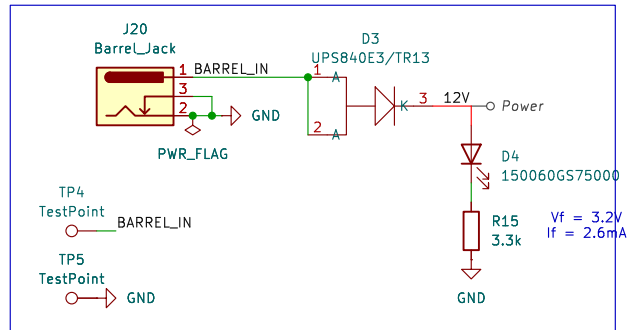
Date: 2024-10-02

Rev: v1.1

KiCad E.D.A. 8.0.5

Id: 7/9

Aux Power In



12V to 5V converter
Max expected current consumption: 2.7 A

Inductor selection math

$$I_{pp} = V_{out}/V_{in(max)} * (V_{in(max)} - V_{out})/(L_o * f_{sw})$$

$$I_{pp} = 5/12.1 * (12.1 - 5)/(3.3uH * 550 \text{ kHz})$$

$$I_{pp} = 1.6 \text{ A}$$

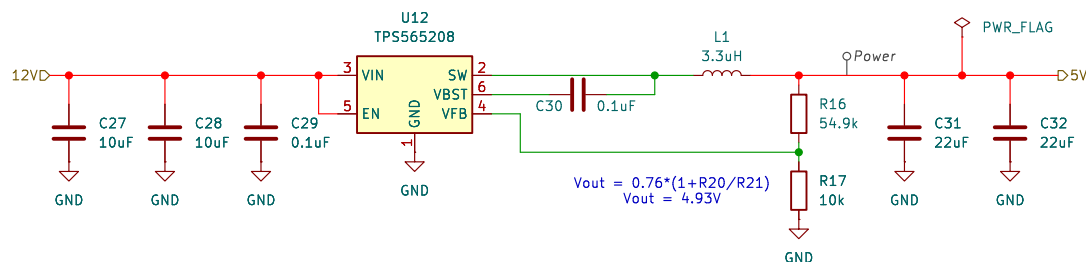
$$I_{peak} = I_o + I_{pp}/2 = 2.7 + 0.8 = 3.5 \text{ A}$$

$$I_o(rms) = \sqrt{I_o^2 + 1/12 I_{pp}^2} = 2.74 \text{ A}$$

Capacitor current

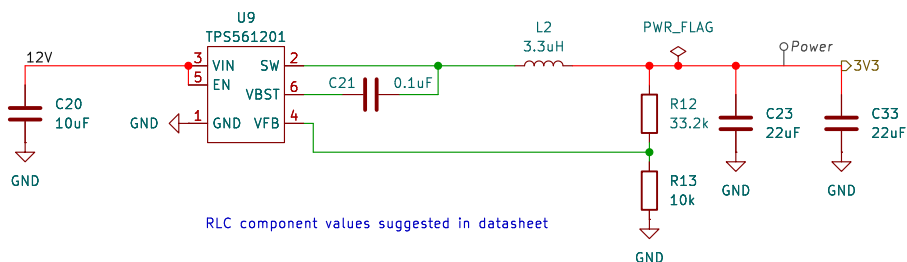
$$I_{co(rms)} = (V_{out} * (V_{in} - V_{out})) / (\sqrt{12} * V_{in} * L * f_{sw})$$

$$I_{co(rms)} = 1.11 \text{ A}$$



12V to 3V3 Converter

Expected current consumption: 126mA
Supports max 500mA

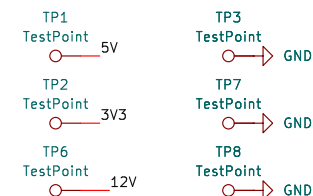


RLC component values suggested in datasheet

$$I_{pp} = 3.3/12.1 * (12.1 - 3.3)/(3.3uH * 580kHz) = 1.25 \text{ A}$$

$$I_{peak} = 0.5 + 1.25/2 = 1.125 \text{ A}$$

$$I_{lo(rms)} = \sqrt{0.5^2 + 1/12 I_{pp}^2} = 0.62 \text{ A}$$



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Sheet: /Power/
File: power.kicad_sch

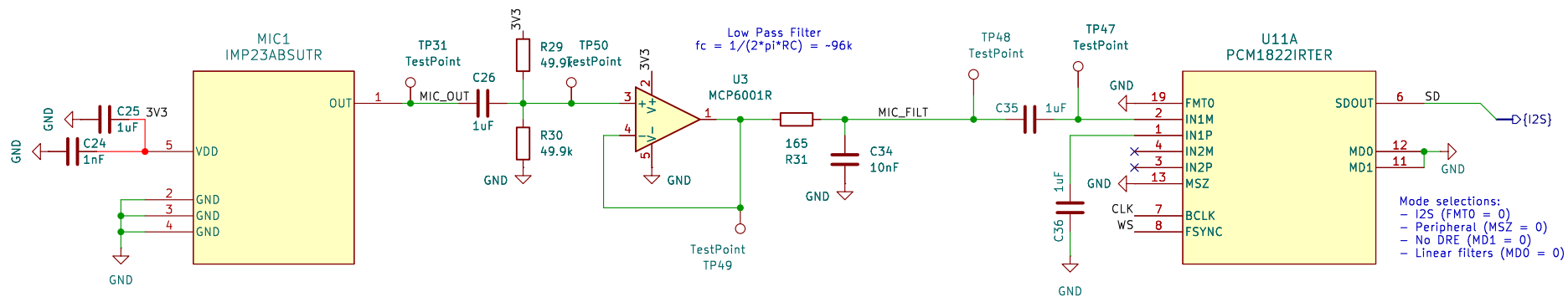
Title: Autotrainer Magnet Module – Power

Size: A4 Date: 2024-10-02

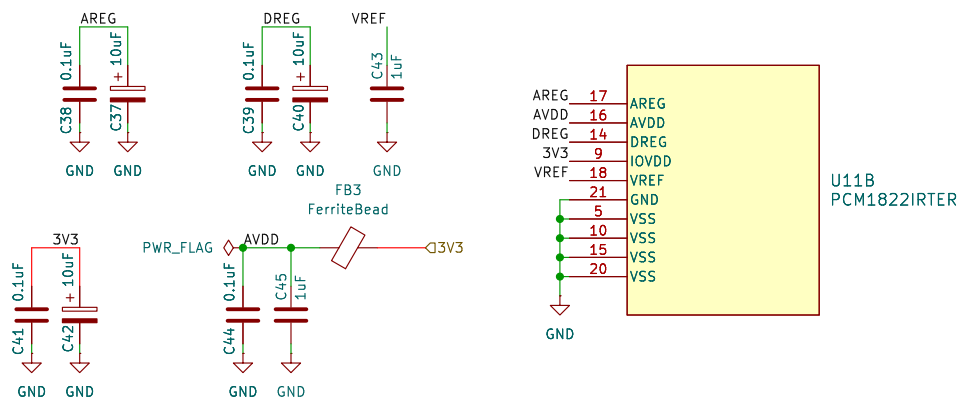
KiCad E.D.A. 8.0.5

Rev: v1.1

Id: 8/9



Decoupling Caps & Power



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Sheet: /Microphone/
File: microphone.kicad_sch

Title:

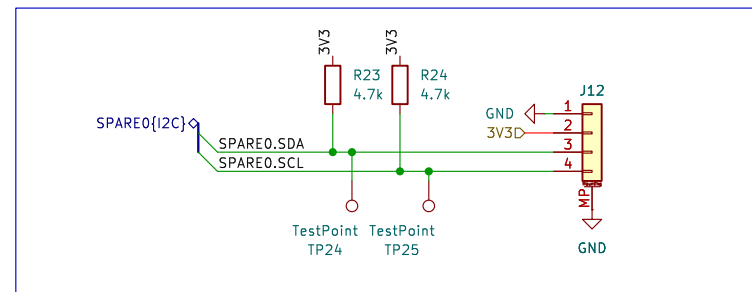
Size: A4	Date: 2024-10-02
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Rev: v1.1

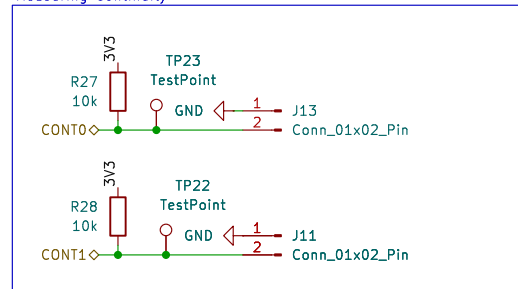
KiCad E.D.A. 8.0.5

Id: 9/9

STEMMA QT Connector



Measuring Continuity



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Sheet: /Extra IO/
 File: io.kicad_sch

Title: Autotrainer Magnet Module – External IO

Size: A4	Date: 2024-10-02	Rev: v1.1
KiCad E.D.A. 8.0.5		Id: 9/9